Faculty of Engineering and Technology
Electrical and Computer Engineering Department

ENCS3340: Artificial Intelligence

Project#1

Optimization Strategies for Local Package Delivery Operations

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Section: 1

Section: 2

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1. Introduction:

The main objective of this project is to design and implement optimization algorithms that can intelligently manage delivery routes and resource allocation. Two heuristics algorithms were chosen for this purpose: Simulated Annealing and Genetic Algorithm. These methods are well-suited for solving complex combination problems with constraints, such as this package-to-vehicle assignment task.

By formulating the problem using Euclidean distance and incorporating soft constraints for priority handling, the system aims to find practical and near-optimal delivery plans. The algorithms allow for some flexibility in priority enforcement when necessary to reduce total travel distance. Each package has a destination (given as coordinates), a weight, and a priority level. Vehicles have limited weight capacities, and the total cost is defined as the total distance traveled by all vehicles.

2. Problem Formulation:

The goal of this project is to optimize the daily delivery plan of a local package delivery service by minimizing the total distance traveled while respecting vehicle capacity constraints and delivery priorities. This problem is modeled as a constraint-based combinatorial optimization task, where each state represents a complete assignment of packages to vehicles and the sequence in which each vehicle delivers its packages.:

State Representation: A state is a complete solution consisting of a list of vehicles, where each vehicle has an ordered list of packages assigned to it. Each package is defined by four values: (x, y) destination, weight, and priority. This representation matches the delivery scenario, as vehicles must follow a route and deliver items in a sequence.

Initial State:The initial state is generated randomly by the function distribute_packages_randomly(), where all packages are assigned to vehicles without exceeding their capacity. This satisfies the requirement from the slides to start with a possibly non-optimal but valid initial guess.

Objective / Fitness Function: As per the local search framework, the fitness function evaluates the quality of a state. In our case, it combines two elements:

- The **total Euclidean distance** traveled by all vehicles from the shop (0,0) to package locations and back.
- A **penalty score** for each pair of packages in a vehicle where a lower-priority package is delivered before a higher-priority one.
  The final fitness value is calculated as:
  Fitness = Total Distance + α × Priority Violations,
  where $\alpha = 10$ is a tuning parameter, as suggested by the project.

Constraints Handling:

Each vehicle must not exceed its weight capacity. This is enforced during both initialization and all optimization steps.

Each package must be assigned to only one vehicle, ensured during crossover, mutation, and initialization.

Successor Function: In Simulated Annealing, successors are generated by selecting two vehicles and swapping one package from each, or reordering packages within a vehicle. A new state is accepted based on whether it improves the fitness or by a probabilistic criterion based on the current temperature. This directly implements the slide's definition of a "local modification to improve the current state." In the Genetic Algorithm, successors are generated through **crossover** and **mutation** operations on parent individuals (solution states), forming the basis of new generations.

Search Space: The number of possible assignments and orderings is extremely large (similar to the TSP and Knapsack examples in the slides), which justifies the use of **Simulated Annealing and Genetic Algorithms**—two standard local search algorithms known for their ability to explore large and complex spaces efficiently without storing search trees.

Termination Condition:

For Simulated Annealing:The process stops when the temperature drops below a threshold ($T < 1$).

For Genetic Algorithm: It stops after a fixed number of generations (500).

3. Heuristics and Constraint Handling:

- **Heuristics in Simulated Annealing**:

New solutions are generated by randomly swapping packages between two vehicles, encouraging gradual improvement and allowing occasional worse moves to escape local optima.

- **Heuristics in Genetic Algorithm**:

Crossover combines parts of two parent solutions, and mutation randomly swaps packages to introduce diversity. A repair step ensures valid assignments.

- **Vehicle Capacity Constraint**:

Before adding any package, its weight is checked to ensure it does not exceed the vehicle's remaining capacity.

- **Unique Package Assignment**:

During crossover and mutation, checks are performed to avoid duplicate or unassigned packages. Any issues are fixed by redistributing packages.

- **Soft Constraint for Priority**:

Priority order is not strictly enforced but included as a penalty in the fitness function to encourage correct sequencing without invalidating solutions.

- **Feasibility Maintained**:

All generated solutions during search are kept valid by integrating constraint checks within each step of the algorithm

4. Parameter Tuning:

Both optimization algorithms used in this project required careful tuning of key parameters to balance performance and solution quality:

- **Simulated Annealing Parameters**:

Initial Temperature: Set to 1000 to allow exploration of the search space with flexibility at the beginning.

Cooling Rate: Set to 0.99 to gradually reduce the acceptance of worse solutions over time.

Stopping Temperature: The algorithm stops when the temperature drops below 1, ensuring controlled termination.

Iterations per Temperature Level: 100 iterations per level to allow enough exploration before cooling.

- **Genetic Algorithm Parameters**:

Population Size: 50 individuals to maintain diversity without overwhelming computation.

Number of Generations: 500 generations to allow sufficient evolution of solutions.

Mutation Rate: 0.05 (5%) to introduce occasional random changes and prevent premature convergence.

These values were selected based on common practices and adjusted through trial runs. The goal was to strike a balance between exploration (searching widely) and exploitation (refining good solutions), ensuring that the algorithms converge to high-quality delivery plans within a reasonable runtime.

5. Test Cases:

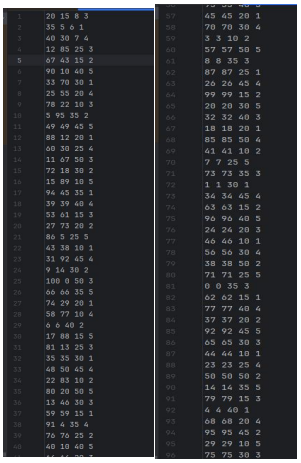

Figure 1Test1:Large input file with 98 packages .

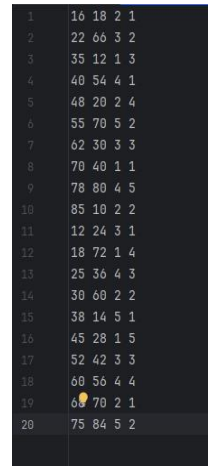

Figure 2Test 2:20 packages in input file

6. Simulated Annealing Testing and Results :

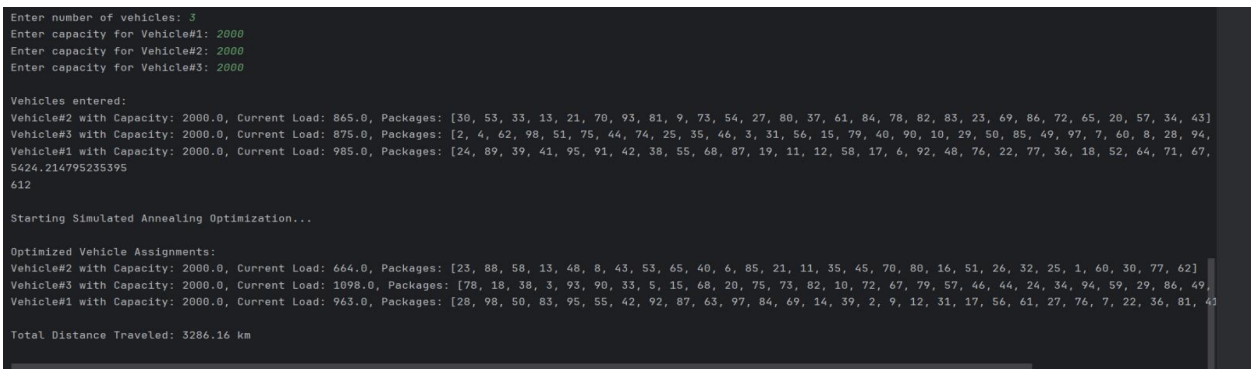

Figure 3:Simulated test1 Output

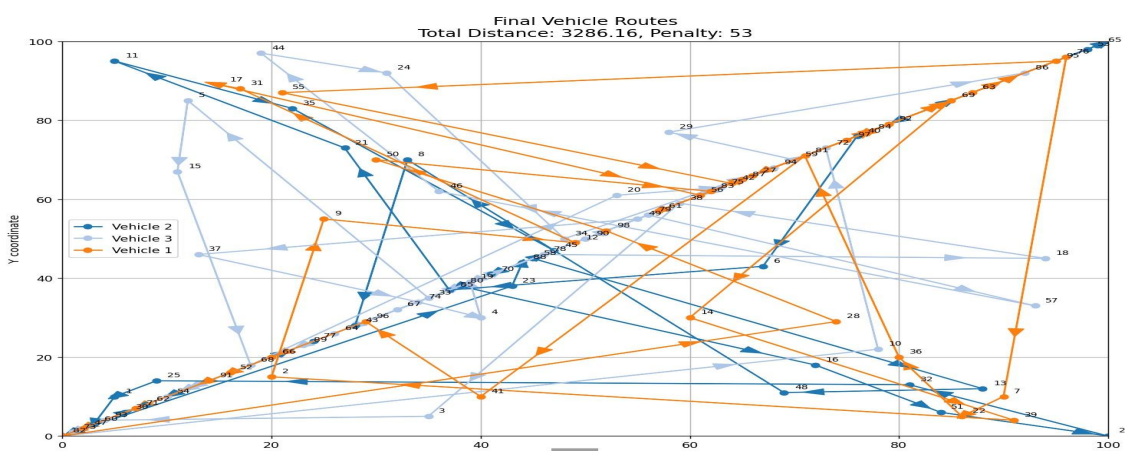

Figure 4 Simulated test 1 Plot

Simulated Ann decreased the distance from 5424 to 3286.16 and penalty from 612 to 53

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C:\Users\bilal\OneDrive\Desktop\AI-Project1\ProjectAi1\.venv\Scripts\python.exe C:\Users\bilal\OneDrive\Deskto

Menu:
1 - Simulated Annealing
2 - Genetic Algorithm
3 - Exit
Choose an option (1-3): 1

Packages loaded successfully

Enter number of vehicles: 2
Enter capacity for Vehicle#1: 100
Enter capacity for Vehicle#2: 100

Vehicles entered:
Vehicle#2 with Capacity: 100.0, Current Load: 30.0, Packages: [5, 18, 14, 20, 6, 17, 19, 13, 2]
Vehicle#1 with Capacity: 100.0, Current Load: 27.0, Packages: [7, 9, 4, 3, 1, 12, 10, 16, 11, 8, 15]
1004.5337081003233
49

Starting Simulated Annealing Optimization...

Optimized Vehicle Assignments:
Vehicle#2 with Capacity: 100.0, Current Load: 27.0, Packages: [19, 20, 6, 10, 7, 17, 14, 13, 3]
Vehicle#1 with Capacity: 100.0, Current Load: 30.0, Packages: [11, 1, 15, 8, 4, 2, 12, 9, 18, 5, 16]

Total Distance Traveled: 716.39 km
```

Figure 5:Simulated test2 Output

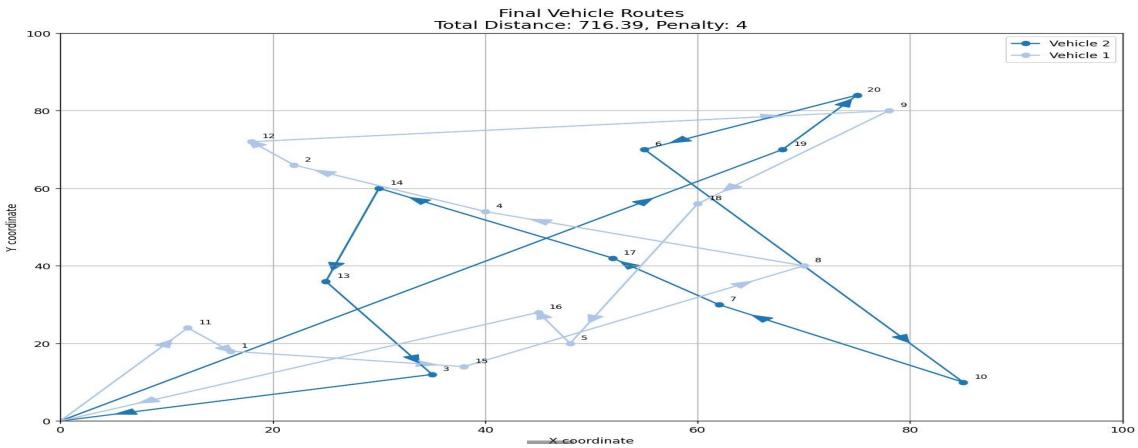

Figure 6 Simulated test2 Output

Simulating annealing decreased distance from 1004 to 716.39 and penalty from 49 to 4.

7. Genetic Algorithm Testing and Results:

```
Enter number of vehicles: 3
Enter capacity for Vehicle#1: 2000
Enter capacity for Vehicle#2: 2000
Enter capacity for Vehicle#3: 2000

Vehicles entered:
Vehicle#3 with Capacity: 2000.0, Current Load: 854.0, Packages: [5, 12, 55, 10, 35, 1, 79, 17, 89, 65, 59, 27, 93, 62, 78, 34, 24, 58, 88, 26, 69, 91, 49, 83, 42, 73, 77, 80, 9
Vehicle#2 with Capacity: 2000.0, Current Load: 1108.0, Packages: [44, 22, 56, 40, 95, 84, 54, 25, 23, 71, 45, 81, 29, 14, 19, 87, 57, 6, 50, 32, 13, 36, 74, 60, 8, 52, 41, 33,
Vehicle#1 with Capacity: 2000.0, Current Load: 763.0, Packages: [38, 75, 96, 70, 48, 82, 51, 28, 47, 63, 98, 72, 67, 61, 53, 90, 64, 4, 66, 94, 76, 18, 85, 97, 21, 31, 3, 16, 4
5223.191830998367
685

Starting Genetic Algorithm Optimization...

Optimized Vehicle Assignments:
Vehicle#1 with Capacity: 2000.0, Current Load: 705.0, Packages: [38, 75, 33, 48, 25, 98, 88, 13, 18, 78, 85, 83, 35, 74, 68, 21, 67, 86, 4, 65, 81, 40, 97, 2, 8, 47, 96, 77, 76
Vehicle#2 with Capacity: 2000.0, Current Load: 844.0, Packages: [30, 80, 70, 58, 37, 5, 72, 90, 82, 1, 55, 64, 87, 95, 92, 53, 6, 11, 59, 51, 9, 41, 60, 44, 57, 62, 36]
Vehicle#3 with Capacity: 2000.0, Current Load: 1176.0, Packages: [73, 93, 43, 23, 3, 16, 63, 50, 20, 32, 49, 42, 94, 28, 45, 69, 31, 79, 17, 89, 34, 71, 54, 7, 26, 27, 61, 84,

Total Distance Traveled: 4637.25 km
```

Figure 7Genatic test1 Output

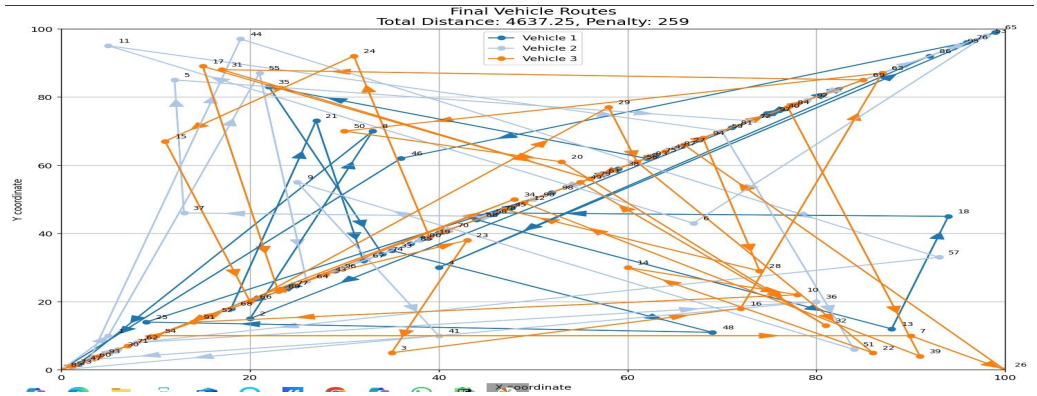

Figure 8 Genatic test1 Plot

Genatic decreased the distance from 5223 to 4637.25 and penalty from 685 to 259

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C:\Users\bilal\OneDrive\Desktop\AI-Project1\ProjectAi1\.venv\Scripts\python.exe C:\Users\bilal\OneDrive\Deskto

Menu:
1 - Simulated Annealing
2 - Genetic Algorithm
3 - Exit
Choose an option (1-3): 1

Packages loaded successfully

Enter number of vehicles: 2
Enter capacity for Vehicle#1: 100
Enter capacity for Vehicle#2: 100

Vehicles entered:
Vehicle#2 with Capacity: 100.0, Current Load: 30.0, Packages: [5, 18, 14, 20, 6, 17, 19, 13, 2]
Vehicle#1 with Capacity: 100.0, Current Load: 27.0, Packages: [7, 9, 4, 3, 1, 12, 10, 16, 11, 8, 15]
1004.5337081003233
49

Starting Simulated Annealing Optimization...

Optimized Vehicle Assignments:
Vehicle#2 with Capacity: 100.0, Current Load: 27.0, Packages: [19, 20, 6, 10, 7, 17, 14, 13, 3]
Vehicle#1 with Capacity: 100.0, Current Load: 30.0, Packages: [11, 1, 15, 8, 4, 2, 12, 9, 18, 5, 16]

Total Distance Traveled: 716.39 km
```

Figure 9 Genatic test 2 Output

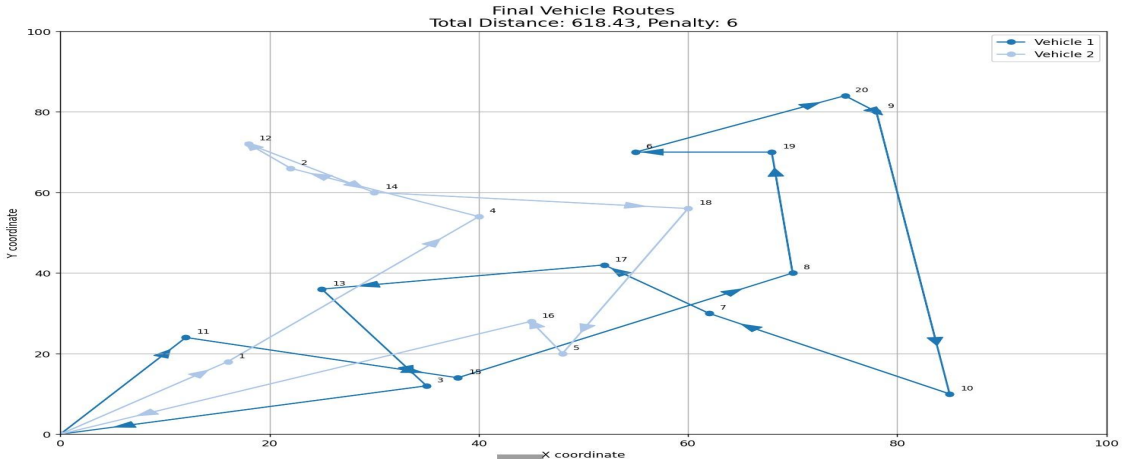

Figure 10 Genatic test2 Output

Genatic decreased the distance from 1004 to 618.43 and penalty from 49 to 6

8. Conclusion

Both Simulated Annealing and Genetic Algorithm are local search techniques that do not store a search tree in memory, thereby significantly reducing space complexity. Instead, they evaluate and compare solutions based on their fitness, guiding the search toward more optimal delivery plans. In our implementation, both algorithms successfully produced near-optimal package-to-vehicle assignments while respecting vehicle capacity constraints and considering package priorities. Simulated Annealing provided strong exploratory capabilities through its temperature-based acceptance mechanism, helping avoid local minima and encouraging diverse solutions. Meanwhile, the Genetic Algorithm maintained a population of candidate solutions, enabling consistent improvement over generations through crossover and mutation. Overall, the system demonstrated efficiency, adaptability, and solution quality, supported by heuristic design and carefully tuned parameters.